

- 2 (a) (i) straight line from $t = 0.60\text{ s}$ to $t = 1.2\text{ s}$ and $|V_v| = 5.9$ at $t = 1.2\text{ s}$
 $V_v = -5.9$ at $t = 1.2\text{ s}$ i.e. line is for negative values of V_v M1
 A1 [2]
- (ii) $s = 0 + \frac{1}{2} \times 9.81 \times (0.6)^2$ or area of graph $= (5.9 \times 0.6) / 2$ C1
 $= 1.8 (1.77)\text{ m}$ $= 1.8 (1.77)\text{ m}$ A1 [2]
- (iii) $V_h = V \cos 60^\circ$ and $V_v = V \sin 60^\circ$ or $V_h = 5.9 / \tan 60^\circ$ or $V_h = 5.9 \tan 30^\circ$ C1
 $V_h = 3.4\text{ m s}^{-1}$ A1 [2]
- (iv) horizontal line at 3.4 from $t = 0$ to $t = 1.2\text{ s}$ [to half a small square] B1 [1]
- (b) (i) $\text{KE} = \frac{1}{2}mv^2$ C1
 $= \frac{1}{2} \times 0.65 \times (6.81)^2$ [allow if valid method to find v] C1
 $= 15 (15.1)\text{ J}$ A1 [3]
- (ii) $\text{PE} = 0.65 \times 9.81 \times 1.77$ C1
 $= 11(11.3)\text{ J}$ A1 [2]

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